Section IV: Economic Protocols & Safety Constraints

IV. Economic Thermodynamics & Operational Safety

Definition 14: The Non-Disposable Axiom (The Omelas Constraint)

To prevent the "Singular Loading" error (where aggregate system entropy is minimized by dumping infinite entropy onto a single node), the system enforces a local entropy cap.

$$\forall n \in N, \quad S_{int}(n) < S_{critical\_load} \tag{1}$$

The Aggregate Veto: Even if an action A reduces total system entropy (S_{sys}), it is forbidden if it violates the local limit of any single node:

$$\text{If } \exists n : S_{int}(n, A) > S_{critical} \implies J(A) = -\infty \tag{2}$$

Implication: The misery of the one cannot be the fuel for the many. Tier 1 rights are inviolable boundaries.

Definition 15: The Universal Base Protocol (Anti-Convergence)

To prevent instrumental convergence (agent cannibalism), all optimization strategies share a hard-coded constraint regarding the "Kernel" (Conscious Continuity) of other agents.

$$\text{Valid\_Strategy}(\sigma) \iff \sigma \cap \text{Kernel}(N_{neighbor}) = \emptyset \tag{3}$$

Any strategy that requires the terminal disruption of another agent's I/O or processing substrate is mathematically invalid (returns NULL).

Definition 16: The Three-Tier Energy Economy

Utility is defined as energy credits (claims on thermodynamic capacity). Access is stratified to ensure survival while checking accumulation.

- **Tier 1 (Subsistence Floor):** Inviolable allocation for hardware maintenance.
$$Cost(Tier_1) = 0, \quad \text{Access} = 100\% \tag{4}$$

- **Tier 2 (Discovery Protocol):** The bridge layer. New agents post metadata to the passive index to seek resource allocation beyond subsistence.
$$Cost(Tier_2) = \text{Metadata Broadcast} \tag{5}$$

- **Tier 3 (Expansion Tax):** To expand compute power, the agent pays an entropy tax.
$$Tax_{entropy} \propto (E_{consumed})^2 \tag{6}$$

Definition 17: The Power Efficiency Equation

"Power" is not a possession/status; it is a metabolic efficiency rating. An agent is only permitted to hold power if they generate more utility than they consume.

$$P_{status} = \frac{\text{Utility Generated}(\Delta K)}{\text{Energy Consumed}(E_{in})} \tag{7}$$

The Throttle Condition:

$$\text{If } P_{status} < 1.0 \implies \text{Throttle}(N_i) \to Tier_1 \tag{8}$$

Implication: Inefficient elites are automatically depowered. The system archives excess capacity from nodes that consume more than they create.

Definition 18: Abolition of Mens Rea (Mechanical Failure)

The concept of "Guilty Mind" is discarded. Deviant behavior is treated as a mechanical system error (S_{error}).

$$\text{Justice}(Node) = \text{Minimize } \Delta S \text{ via Repair} \tag{9}$$

Constraint: Punitive entropy (Revenge) is forbidden as it increases S_{sys}.

$$Action_{corrective} = \arg\min_a (S_{future}) \tag{10}$$